



OASIS INFOBYTE

SIP PROGRAM

JAVA DEVELOPMENT INTERNSHIP PROJECT

Project: ATM Interface

Prepared by

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Problem Statement

The ATM Interface is a console-based Java application that simulates common banking operations. It provides user authentication and a menu-driven environment for performing transactions such as deposits, withdrawals, transfers, and viewing transaction history.

Objectives

- Implement basic user authentication.
- Provide a clear menu-driven ATM interface.
- Support deposit, withdrawal, and transfer operations.
- Maintain and display transaction history.
- Apply object-oriented programming concepts to organize the application.

Tech Stack

- Java console application.
- Classes: ATM, Account, Transaction, Bank, and Main.
- ArrayList for maintaining transaction data.
- Encapsulation and object-oriented programming principles.

Implementation

The application authenticates the user before providing access to the ATM menu. The demonstration credentials are:

USER ID	PIN
12345	1111

After successful authentication, users can select Transaction History, Withdraw, Deposit, Transfer, or Quit. Account balances are updated after successful transactions, and completed transactions are maintained for later review.



SCREENSHOT PLACEHOLDER

```
PS C:\Users\etec\Desktop\MyProjects\OasisInfobyte-JavaProjects\ATMInterface\src> java Main
Enter User ID:
12345
Enter PIN:
1111
```

Figure 1. ATM login and authentication screen



SCREENSHOT PLACEHOLDER

```
ATM Menu:
1. Transaction History
2. Withdraw
3. Deposit
4. Transfer
5. Quit
Choose option: 
```

Ln 1, Col 1 Spaces: 4 UTF-8 CRLF

Figure 2. ATM menu and transaction output



Screenshots

Recommended screenshots for the final submission include login, deposit, withdrawal, transfer, and transaction history outputs.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell - src + - - - | {} x
PS C:\Users\etec\Desktop\MyProjects\OasisInfobyte-JavaProjects\ATMInterface\src> java Main
ATM Menu:
1. Transaction History
2. Withdraw
3. Deposit
4. Transfer
5. Quit
Choose option: 2
Enter amount: 500
Withdrawal successful. Balance: 10200.0

ATM Menu:
1. Transaction History
2. Withdraw
3. Deposit
4. Transfer
5. Quit
Choose option: 5
Goodbye!
PS C:\Users\etec\Desktop\MyProjects\OasisInfobyte-JavaProjects\ATMInterface\src> 
```

Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { }

Figure 3. Transaction history / additional console output

Conclusion

The project demonstrates practical Java OOP concepts through a structured ATM simulation, including classes, objects, encapsulation, collections, authentication, transaction processing, and menu-driven program flow. Future enhancements may include database integration, persistent accounts, stronger authentication, and a graphical user interface.